

app.py Code Explanation (Line by Line Details)

1. Imports & Setup (Lines 1-17)

Code Streamlit, Pandas, Matplotlib aur Seaborn ko import karta hai. `plt.style.use('default')` chart ka background white rakhta hai. `st.set_page_config()` website ka title "Sales Analytics Dashboard" aur layout ko "wide" (poori screen width) banata hai.

2. Custom CSS Injection (Lines 19-45)

`st.markdown(..., unsafe_allow_html=True)` ka use karke custom CSS add ki gayi hai. Ye CSS `.kpi-card` class banati hai jisme white background, grey border aur black text hota hai. Ye cards screen par top metrics dikhane ke kaam aate hain.

3. Data Loading - `load_data()` (Lines 47-81)

Ye dashboard ka sabse important function hai. Iske upar `@st.cache_data` likha hai, jiska matlab ek baar data load hone ke baad RAM me save ho jayega (baar baar CSV read nahi hogi). Ye Backend/Processed/ folder se teeno clean CSV files (customers, orders, products) uthata hai. Agar file nahi milti to `st.error()` show karta hai. `OrderDate` ko proper Python Datetime me convert kiya jata hai taaki date filters smoothly chal sakein.

4. Helper Functions (Lines 83-96)

- `format_currency(value)`: Ye raw data (jaise 2500000) ko padhne layak Indian format (jaise Rs 25.00 L ya Rs 2.50 Cr) me badalta hai.
- `format_number(value)`: Simple numbers mein commas lagata hai (jaise 10000 ko 10,000 banata hai).

5. Data Merging / Joining (Lines 98-103)

Pandas ka `.merge()` method SQL ke JOIN ki tarah kaam karta hai. `orders_df` ke andar `customers_df` ko (CustomerID se) aur `products_df` ko (ProductID se) jod diya jata hai. Isse ek badi "merged_df" banti hai jisme transaction ke sath Customer ki City aur Product ki Category bhi hoti hai.

6. Sidebar & Filters (Lines 105-170)

`st.sidebar` dashboard ke left side me menu banata hai. Isme "Cascading Filters" lagaye gaye hain. Matlab agar aap Year = 2025 chunte ho, to agla filter (Month) sirf unhi mahinon ke naam dikhayega jo 2025 me the. Date Range picker (`st.date_input`) dynamically bache hue data ki min aur max date nikalta hai aur us range ke bahar ka data filter kar deta hai (`start_date <= Date <= end_date`).

7. KPI Calculations (Lines 172-188)

`st.columns(7)` screen ko 7 barabar hisso me todta hai. Uske baad `filtered_df` par calculations hoti hain:

- `.nunique()` se Total Orders aur Customers nikalte hain.
- `.sum()` se Total Products aur Revenue nikalta hai.
- `.mean()` se Average Order Value nikalta hai.

Fir in values ko `.metric()` use karke pehle banaye gaye CSS styled cards me dikhaya jata hai.

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8. Dashboard Tabs (Lines 191 to End)

st.tabs() se 6 alag-alag tabs banaye gaye hain, taki page bahut lamba na ho:

- Overview (Tab 1): `groupby().sum().idxmax()` ka use karke automatically highest revenue category aur top city khoji jati hai (hardcode nahi hai). Phir `sns.heatmap()` numeric columns ka correlation dikhata hai.
- Sales Analysis (Tab 2): Data ko Year aur Month se group karke `ax.plot()` se Line Chart banaya gaya hai. `.shift(1)` ka use pichle mahine ki sales nikalne aur Growth % calculate karne ke liye hota hai.
- Product & Customer (Tab 3 & 4): `sns.barplot()` ka use karke Top 10 products aur top customers ka bar chart draw hota hai. Sath hi Customer Segmentation dikhaya gaya hai.
- City & Payments (Tab 5 & 6): `ax.pie()` ka use karke Order Status aur Payment method ka Pie Chart banaya hai. Jisme % show hota hai.

Har chart ke baad `st.dataframe()` function data ko ek interactive, scroll hone wali table me dikha deta hai.